

## APPM 4600 — HOMEWORK # 2

1. (a) Show that  $x \sin \sqrt{x} = O(x^{3/2})$  as  $x \rightarrow 0$ .  
 (b) Show that  $\int_0^\varepsilon e^{-x^2} dx = O(\varepsilon)$  as  $\varepsilon \rightarrow 0$ .
  
2. Consider solving  $\mathbf{Ax} = \mathbf{b}$  where  $\mathbf{A} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 + 10^{-10} & 1 - 10^{-10} \end{bmatrix}$  and  $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ . The exact solution is  $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and the inverse of  $\mathbf{A}$  is  $\begin{bmatrix} 1 - 10^{10} & 10^{10} \\ 1 + 10^{10} & -10^{10} \end{bmatrix}$ . In this problem we will investigate a perturbation in  $\mathbf{b}$  of  $\begin{bmatrix} \Delta b_1 \\ \Delta b_2 \end{bmatrix}$  and the numerical effects of the condition number.
  - (a) Find an exact formula for the change in the solution between the exact problem and the perturbed problem  $\Delta x$ .
  - (b) What is the condition number of  $\mathbf{A}$ ?
  - (c) Let  $\Delta b_1$  and  $\Delta b_2$  be of magnitude  $10^{-5}$ ; not necessarily the same value. What is the relative error in the solution? What is the relationship between the relative error, the condition number, and the perturbation. Is the behavior different if the perturbations are the same? Which is more realistic: same value of perturbation or different value of perturbation?
  
3. Let  $f(x) = e^x - 1$ 
  - (a) What is the relative condition number  $\kappa(f(x))$ ? Are there any values of  $x$  for which this is ill-conditioned?
  - (b) Consider computing  $f(x)$  via the following algorithm:
 

```
1: y = math.e^x
2: return y - 1
```

Is this algorithm stable? Justify your answer
  - (c) Let  $x$  have the value  $9.999999995000000 \times 10^{-10}$ , in which case  $f(x)$  is equal to  $10^{-9}$  up to 16 decimal places. How many correct digits does the algorithm listed above give you? Is this expected?
  - (d) Find a polynomial approximation of  $f(x)$  that is accurate to 16 digits for  $x = 9.999999995000000 \times 10^{-10}$ . Hint: use Taylor series, and remember that 16 digits of accuracy is a relative error, not an absolute one.
  - (e) Verify that your answer from part (d) is correct.
  - (f) [Optional] How many digits of precision do you have if you do a simpler Taylor series?
  - (g) [Fact; no work required] Matlab provides `expm1` and Python provides `numpy.expm1` which are special-purpose algorithms to compute  $e^x - 1$  for  $x \approx 0$ . You could compare your Taylor series approximation with `expm1`.
  
4. Consider the equation  $2x - 1 = \sin x$ .

- (a) Find a closed interval  $[a, b]$  on which the equation has a root  $r$ , and use the Intermediate Value Theorem to prove that  $r$  exists.
  - (b) Prove that  $r$  from (a) is the only root of the equation (on all of  $\mathbb{R}$ ).
  - (c) Use the bisection code from class (or your own) to approximate  $r$  to eight correct decimal places. Include the calling script, the resulting final approximation, and the total number of iterations used.
5. (a) Use a theorem from class (Theorem 2.1 from text) to find an upper bound on the number of iterations in the bisection needed to approximate the solution of  $x^3 + x - 4 = 0$  lying in the interval  $[1, 4]$  with an accuracy of  $10^{-3}$ .
- (b) Find an approximation of the root using the bisection code from class to this degree of accuracy. How does the number of iterations compare with the upper bound you found in part (a)?
6. All the roots of the scalar equation

$$x - 4 \sin(2x) - 3 = 0,$$

are to be determined with at least 10 accurate digits<sup>1</sup>.

- (a) Plot  $f(x) = x - 4 \sin(2x) - 3$  (using your `Python` toolbox). All the zero crossings should be in the plot. How many are there?
- (b) Write a program or use the code from class to compute the roots using the fixed point iteration

$$x_{n+1} = -\sin(2x_n) + 5x_n/4 - 3/4.$$

Use a stopping criterium that gives an answer with ten correct digits. (*Hint: you may have to change the error used in determining the stopping criterion.*) Find, empirically which of the roots that can be found with the above iteration. Give a theoretical explanation.

### Extra fun!

1. The function  $f(x) = (x - 5)^9$  has a root (with multiplicity 9) at  $x = 5$  and is monotonically increasing (decreasing) for  $x > 5$  ( $x < 5$ ) and should thus be a suitable candidate for your function above. Use `a=4.82` and `b=5.2` and `tol = 1e-4` and use `bisection` with:
  - (a)  $f(x) = (x - 5)^9$ .
  - (b) The expanded expanded version of  $(x - 5)^9$ , that is,  $f(x) = x^9 - 45x^8 + \dots - 1953125$ .
  - (c) Explain what is happening.

### 2. Practicing Python

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<sup>1</sup> $n$  accurate digits is equivalent to a relative error smaller than  $0.5 \times 10^{-n}$ .

- (a) Create a vector  $\mathbf{t}$  with entries starting at 0 incrementing by  $\frac{\pi}{30}$  to  $\pi$ . Then create the vector  $\mathbf{y} = \cos(\mathbf{t})$ .

Write a code that evaluates the following sum:

$$S = \sum_{k=1}^N \mathbf{t}(k) \mathbf{y}(k)$$

Print the statement “the sum is:  $S$ ” with the numerical value of  $S$ .

- (b) *Wavy circles*. In one figure, plot the parametric curve

$$\begin{aligned} x(\theta) &= R(1 + \delta r \sin(f\theta + p)) \cos(\theta) \\ y(\theta) &= R(1 + \delta r \sin(f\theta + p)) \sin(\theta) \end{aligned}$$

for  $0 \leq \theta \leq 2\pi$  and for  $R = 1.2$ ,  $\delta r = 0.1$ ,  $f = 15$  and  $p = 0$ . Make sure to adjust the scale so that the axis have the same scale.

In a second figure, use a *for* loop to plot 10 curves and let with  $R = i$ ,  $\delta r = 0.05$ ,  $f = 2 + i$  for the  $i^{th}$  curve. Let the value of  $p$  be a uniformly distributed random number (look up `random.uniform`) between 0 and 2.